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A novel *in-vitro* model of the bone marrow microenvironment in AML identifies CD44 and Focal Adhesion Kinase as therapeutic targets to reverse cell adhesion-mediated drug resistance

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Abstract

Acute myeloid leukemia (AML) is an aggressive neoplasm. Although most patients respond to induction therapy, they commonly relapse due to recurrent disease in the bone marrow microenvironment (BMME). So, disruption of the BMME, releasing tumour cells into the peripheral circulation, has therapeutic potential. Using both primary donor AML cells and cell lines, we developed an *in-vitro* co-culture model of the AML BMME. We used this model to identify the most effective agent(s) to block AML cell adherence and reverse adhesion-mediated treatment resistance. We identified anti-CD44 treatment significantly increased the efficacy of cytarabine. However, some AML cells remained adhered, and transcriptional analysis identified focal adhesion kinase (FAK) signalling as a contributing factor; adhered cells showed elevated FAK phosphorylation that was reduced by the FAK inhibitor, defactinib. Importantly, we demonstrated that anti-CD44 and defactinib were highly synergistic at diminishing adhesion of the most primitive $CD34^{high}$ AML cells in primary autologous co-cultures. Taken together, we identified anti-CD44 and defactinib as a promising therapeutic combination to release AML cells from the chemoprotective AML BMME. As anti-CD44 is already available as a recombinant humanised monoclonal antibody, the combination of this agent with defactinib could be rapidly tested in AML clinical trials.

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Introduction

AML is characterised by uncontrolled expansion of myeloid progenitors in the bone marrow (BM) and peripheral blood (PB) and remains a therapeutic challenge due to its aggressive nature and genetic and phenotypic heterogeneity[1,2]. Relapse after initial response to chemotherapy occurs in the majority of patients (~80%)[3] and remains a serious clinical challenge requiring new therapeutic strategies. The estimated median overall survival (OS) of AML is 8.5 months, with a 24% 5-year OS. Nearly 80% of AML patients diagnosed at ≥65 years will die within one year[4]. One of the issues hindering successful treatment of AML, and contributing to disease relapse, is leukemic cell retention in the protective niche of the BM[5]. Here, the leukemic cells are surrounded by multiple cell types that promote tumor cell survival, enabling them to evade destruction by systemic therapies, leading to the emergence of drug resistance[6].

The mechanisms that lead to adhesion-mediated protection of leukemic cells in the BM are complex and involve multiple cytokines, chemokines and adhesion molecules. Three cell types that have been identified as important components of the BMME are stromal cells, endothelial cells and osteoblasts[7,8]. Stromal cells have been shown to protect AML cells from spontaneous and drug-induced apoptosis by direct contact[5], and reciprocal NF-κB activation[9]. They induce NF-κB signaling which leads to the transcriptional activation of anti-apoptotic genes in AML cells resulting in increased tumor cell survival and resistance to chemotherapy[10]. Stromal cells are also known to enhance the metabolic activity of AML cells which contributes to the chemo-resistance associated with residence in the BMME[11]. Endothelial cells not only provide an essential nutrient-delivering vascular network for the tumor cells, but also play a role in tumor cell adhesion and migration between the BM and circulation[12]. Signaling

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between leukemic and endothelial cells is bi-directional as leukemia-mediated endothelial cell activation has been found to upregulate the cell adhesion molecule E-selectin, promoting AML cell quiescence and a chemoresistant state[13]. The presence of osteoblasts also creates a tumor-promoting niche; AML cells can promote osteogenic differentiation of stromal cells into osteoblasts[14]. Despite all three cell types having tumor promoting properties, drugs that target these interactions are not routinely used in AML.

AML and normal hemopoietic stem cells (HSCs) adhere to the BM via a number of mechanisms including the receptors CXCR4, very late antigen-4 (VLA-4) and the cell surface glycoprotein CD44[15]. Elevated expression of CXCR4 on AML cells is associated with decreased patient survival[16] and attempts have been made to mobilize leukemic cells out of the protective niche of the BMME using CXCR4 inhibitors such as plerixafor. Clinical trials demonstrated some efficacy of CXCR4 inhibitors, however not all AML cells are mobilized[17,18]. This indicates that, whilst the CXCR4 interaction with its ligand CXCL12 is important, other molecules play a role in adhesion of AML cells in the protective niche of the BMME. VLA-4, is an integrin dimer composed of CD49d (α4) and CD29 (β1) and has also been associated with chemoresistance in AML[19]. VLA-4 binds to the extracellular matrix (ECM) protein fibronectin and VCAM-1 expressed by stromal and endothelial cells. In a murine model of AML, survival was significantly prolonged in mice that were given VLA-4 blocking antibodies in combination with cytarabine than those receiving cytarabine alone[19]. CD44 is a glycosylated class-I transmembrane protein that binds to the glycosaminoglycan, hyaluronic acid (HA; an important component of the BM ECM) [20], osteopontin (OPN), collagens, and matrix metalloproteinase[20]. HA and CD44 expression have been

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implicated in AML, with Hartmann *et al*. demonstrating the importance of CD44-induced activation of VLA-4 in AML adhesion[21]. Expression of CD44 is essential for AML engraftment in a mouse model[22] and targeting CD44 reduces AML stem cell re-population in a serial transplant model[23]. Interestingly, CD44 has been shown to be a key player in drug resistance of other haematological malignancies such as multiple myeloma[24,25]. To target BM resident chemo-resistant AML cells, a better understanding of the biological interactions between leukemic cells and the hemopoietic niche is needed. In view of the above, CXCR4, VLA-4, CD44 and E-selectin represent promising targets for further investigation.

The aims of this study were two-fold. Firstly, to develop a multi-cellular *in-vitro* co-culture system that mimics the adhesive AML-BMME. Secondly, to identify the most effective agent(s) for blocking adhesion of AML cells and to establish whether they can reverse cell adhesion-mediated cytarabine resistance.

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Methods

Full details of all methods are provided in the supplementary methods

Cell culture

Cell lines: KG1a, OCI-AML3, HS-5, hFOB 1.19 and HUVECs were purchased from ATCC (USA) and DSMZ (Germany) and grown in the recommended media. hTERT transfected HUVECs were previously generated and characterised over multiple passages in the Pepper lab[26].

Primary cells: Patients with a new AML diagnosis were recruited (supplementary Table 1). PB samples and/or BM samples were obtained with full ethical approval and following informed consent in accordance with the Declaration of Helsinki.

Co-culture and cell adhesion quantification: Co-culture assays were performed on 12, 24 and 96-well plates with consistent cell density and stroma:AML cell ratio. Full details of this, drugs used, isolation/identification and counting of non-adhered and adhered AML cells are in the supplementary methods and supplementary Table 2.

Immunophenotyping

Labeling with fluorescent antibodies was carried out according to the manufacturer's instructions using the clones listed in supplementary Table 3. The expression of AML markers were quantified on $CD45^{high}CD73^{low}$ AML cells (stromal cells were all $CD45^{low}CD73^{high}$) using median fluorescent intensity (MFI). All immunophenotyping data were acquired on a CytoFLEX LX flow cytometer (Beckman Coulter).

AML cell isolation

The EasySep™ human CD45 Positive Selection Kit and an EasySep™ magnet were used to isolate the AML cells as per the manufacturer's instructions. To ensure all AML cells were treated in an identical fashion, both adhered and non-adhered AML cells were selected.

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Measurement of *in-vitro* apoptosis

Selected adhered, non-adhered and monoculture AML cells were stained using the BioLegend Annexin V FITC kit (Biolegend) following the manufacturer's instructions.

RNA sequencing and statistical analysis including synergy

Full details are provided in the supplementary methods.

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Results

The 'BM adhesion system' (BMAS)

Although many *in-vitro* co-culture systems have been established for AML, none have measured AML cell adhesion in a multi-cellular, bone marrow-relevant, system. Stromal cells, endothelial cells and osteoblasts are abundant and importantly, functionally significant in the BMME[7,8], potentially all using different cocktails of adhesion receptor/ligands for adhesion. Therefore, to create a model that mimics the heterogeneous adhesive and protective BMME, and which can be used as a drug testing platform, we used HS-5 (stromal cells), HUVEC (endothelial cells) and hFOB 1.19 (osteoblasts). We found all three to be adhesive to AML cells to variable degrees, (supplementary Figure 1) with different expression levels of common adhesion markers (supplementary table 3; e.g. HUVECs are 100% CXCR4 positive whereas HS5s don't express CXCR4) highlighting the importance of using a multicellular system. Subsequently, we used HS5, hFOB 1.19 and HUVEC (stroma mix) plated in equal proportions (1:1:1; the BMAS)(supplementary Figure 1).

AML cell lines were chosen to represent distinct AML phenotypes. KG1a cells are BM-derived AML cells representing a more primitive AML disease (containing a $CD34^{hi}CD38^{dim}$ stem like sub-population) [27]. In contrast, OCI-AML3 cells are PB-derived and represent a more differentiated disease[28] (supplementary Table 3). Despite their divergent characteristics, both AML cell lines exhibited strong adhesion on the BMAS in comparison to those cultured alone (KG1a non-adhered; 23%, p=0.0019 and OCI-AML3 non-adhered; 28%, p=0.004) (Figure 1a-b).

To confirm that adherence in the BMAS mimics the protective nature of the BMME, we measured the viability BMAS-adhered and non-adhered AML cells. The adhered cells were both significantly less apoptotic (OCI-AML3; 25% apoptotic versus 42%,

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p=0.001, KG1a; 6.8% apoptotic versus 14%, p=0.001 in adhered versus non-adhered respectively; Figure 1c). These results confirm that the *in-vitro* BMAS recapitulates some of the functionally important aspects of the AML-BMME, namely its adhesive nature and promotion of tumor cell viability.

The BMAS models cell adhesion-mediated drug resistance (CAM-DR) and soluble factor-mediated drug resistance (SFM-DR).

To establish whether the BMAS also models AML chemo-resistance, we performed cytarabine dose-response assays and measured apoptosis with Annexin V after 72 hours. Strikingly, in both OCI-AML3 and KG1a cells, the LC_{50} (cytarabine dose that kills 50% of cells) was not achieved in the adhered AML cells even at the highest dose of 10μM, indicating a strong chemotherapy resistance effect created by adhesion in the BMAS; adhered LC_{50} >10μM. For the BMAS non-adhered the LC_{50} values were OCI-AML3 7.38μM and KG1a 4.57μM and for monoculture cells OCI-AML3 1.87μM and KG1a 2.81μM (Figure 2). As the non-adhered cell LC_{50} was higher than monoculture, soluble factors in the BMAS must also contribute to cytarabine resistance, but not as much as cell adhesion (Figure 2). Importantly, cytarabine had no significant effect on the stroma mix cells (Figure 2).

Our *in-vitro* system therefore successfully emulates both the cell adhesion mediated drug resistance (CAM-DR) and soluble factor mediated drug resistance (SFM-DR), seen in the BMME. These results also suggest that prevention of AML adhesion within the BMME could diminish chemotherapy resistance.

Anti-CD44 most effectively reduces AML adhesion.

We next sought to see if we could prevent AML cell adhesion in the BMAS. Our phenotyping studies identified that both AML cell lines expressed high levels of the

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adhesion molecules CXCR4, CD49d and CD44 (supplementary Table 3). Therefore, we tested whether adhesion could be prevented using several commercially available blocking agents to CXCR4 (plerixafor[17,18] and ONO-7161), CD49d (natalizumab) and CD44 (anti-CD44). We also tested an E-selectin blocking antibody as it is expressed on the 21% of HUVEC cells in our co-culture (supplementary Table 3) and has previously been suggested as important in AML cell adhesion[13,29-31].

Testing the impact of each blocking agent across a range of concentrations revealed that treatment with anti-CD44 significantly reduced adhesion at all doses in both cell lines (Figure 3a and b). Compared to untreated controls, 5 µg/mL anti-CD44 treatment resulted in a 1.8-fold reduction in adhesion of OCI-AML3 cells (Figure 3a), and a 3.6-fold reduction in adhesion in KG1a cells (Figure 3b). Importantly, this concentration of anti-CD44 was more effective than the highest concentration of CXCR4 blocking agents (plerixafor and ONO-7161), CD49d (natalizumab) and E-selectin (anti-E-selectin) (Figure 3c). Individual dose-response plots for each of these agents can be seen in supplementary Figure 2.

We next tested the effectiveness of anti-CD44 on primary BM and PB derived AML cells. Compared to untreated, anti-CD44 reduced adhesion of both BM and PB derived AML cells up to 4-fold with a mean of 1.3-fold ($p<0.001$, $n=10$) and 1.5-fold ($p<0.0001$, $n=15$), respectively (Figure 3d). In summary, our co-culture system revealed that blocking CD44 was effective at blocking AML cell adhesion in both cell lines and primary cells. We hypothesised that if CD44 is an important adhesion marker for BM retention, BM expression of CD44 will negatively correlate with the PB white blood cell (WBC) count as AML cell retention in the BM will reduce AML cells in the PB. Using 8 BM-derived primary samples, the surface expression of adhesion molecules on blasts was assessed for correlation with the PB WBC count (taken at

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the time of BM biopsy). The PB WBC count was negatively correlated with CD44 expression (r=-0.7, p=0.044; Figure 3e). In contrast, no correlation was observed with CXCR4 (r=-0.23, p=0.584) or CD49d (r=0.007, p=0.987), supporting the hypothesis that CD44 is one of the most important molecules in BM retention of AML cells. To confirm this, we compared previously published BM CD44 mRNA levels with the percentage of AML blasts in the BM and found a significant positive correlation (r=0.32, p<0.0001; Figure 3f).

Anti-CD44 treatment potentiates anti-tumor effects by reducing CAM-DR

We next combined anti-CD44 with chemotherapy in our *in-vitro* culture system to investigate whether modulating adhesion through CD44 can reduce CAM-DR. Compared to no antibody controls, the addition of 5µg/mL anti-CD44 resulted in significantly more apoptotic AML cells using three different cytarabine concentrations (Figure 4a); 1.39-1.66-fold increase in apoptotic OCI-AML3 cells and a 1.44-1.8-fold increase in apoptotic KG1a cells. A representative dot plot is shown in Figure 4b, where the combination treatment resulted in more non-adhered AML cells and therefore more Annexin V^+ apoptotic AML cells. Figure 4c shows the residual co-culture after all the non-adhered cells were removed and reveals that the combination resulted in strikingly few viable AML cells still adhered to the stroma mix which itself remained viable and adhered to the plastic plate.

These findings were confirmed using primary AML cells where combination treatment resulted in up to 1.9-fold increase in apoptotic cells compared to cytarabine alone in both PB and BM derived AML cells (Figure 4d and 4e). Interestingly, we found that for 13/15 individual patient samples, more apoptosis was seen in the wells containing both drugs (5µg/mL anti-CD44 with 5µM cytarabine) than when we added together the apoptosis seen in the wells with cytarabine alone with that in anti-CD44 alone;

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mean±SD apoptotic cell number 3036±2482 [combination] versus 1928±1772 [sum] p=0.009 (supplementary Figure 4a and Figure 4f). Importantly, in the presence of anti-CD44 and cytarabine, the number of apoptotic primary AML cells correlated with their pre-experiment expression of CD44 (r=0.73, p=0.002; supplementary Figure 4b).

Transcriptomic analysis identified FAK as a determinant of persistent adhesion following treatment with anti-CD44

Despite anti-CD44 treatment being the most effective at preventing AML adhesion, some AML cells (using both cell lines and primary AML cells) remained persistently adhered and viable, even in the presence of the highest dose of anti-CD44 and high (micromolar) concentrations of cytarabine. This reflects the heterogeneity of AML, even within a cell line. To identify novel adhesion targets, we isolated and performed RNA-sequencing (RNA-seq) on persistently adhered cells and non-adhered cells after anti-CD44 treatment. Unsupervised clustering of RNA-seq data showed a transcriptional signature associated with persistent adhesion which was largely consistent across cell lines (Figure 5a). Differential expression analysis was performed separately for each cell line and identified 215 common differentially upregulated genes in the adhered cells compared to non-adhered (Figure 5b). Most differentially expressed genes were upregulated (486 for OCI-AML3 and 364 for KG1a), with only a few genes showing downregulation (Figure 5c). KEGG pathway enrichment analysis identified over-representation of the focal adhesion kinase (FAK) signalling pathway in adhered cells of both cell lines, suggesting that targeting this pathway may be therapeutically beneficial for overcoming persistent adhesion in the presence of anti-CD44 (Figure 5d). The RNA-seq results were validated with qRT-PCR (supplementary

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Figure 5) and uploaded to the EMBL-EBI data repository (accession number E-MTAB-14052).

While the expression of FAK itself remained unchanged, measuring FAK phosphorylation (pFAK) in BMAS in the presence and absence of anti-CD44 showed an increase in FAK signaling in adhered versus non-adhered AML cells. As before, the number of adhered AML cells was much less in the presence of anti-CD44 but pFAK was consistently higher in the adhered AML cells (MFI 3133±94) compared to non-adhered (MFI 2052±34) and monoculture (MFI 1661±99) supporting their use of this signalling pathway for adhesion (Figure 5e).

Defactinib potentiates anti-CD44 treatment in preventing AML cell adhesion to the BMME.

As the transcriptomic analysis highlighted the FAK pathway as a potential mechanism of AML adhesion, we attempted to overcome persistent adherence by targeting this pathway in combination with anti-CD44.

Defactinib, a potent dual and reversible ATP-competitive inhibitor of FAK and PYK2[32], is currently in phase II clinical trials for the treatment of several solid tumors[33,34]. We initially tested the potential of defactinib to reduce pFAK in AML cells and found that after an 18-hour incubation followed by 3 hours on the BMAS (with and without anti-CD44), defactinib reduced pFAK in both adhered and non-adhered AML cells ($p=0.039$; Figure 5f). We next tested defactinib as an anti-adhesion therapy in our model and compared to untreated, found that it significantly reduced adhesion of the KG1a cell line (mean reduction of 1.5-fold; $p<0.04$) (supplementary Figure 6a). Surprisingly, defactinib did not have the same effect on the OCI-AML3 cell line (supplementary Figure 6b). Although elements of the FAK pathway were over-

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expressed in adhered cells from both cell lines, expression of FAK (PTK2) itself was five times higher in the more primitive KG1a cells compared to OCI-AML3 cells, which may explain their increased sensitivity to defactinib. To investigate whether higher expression of FAK was associated with a more primitive phenotype in AML, we compared previously published CD34 and FAK (PTK2) mRNA levels in AML cell lines and two AML patient cohorts[35-37]. Despite the challenges of patient-to-patient heterogeneity across multiple RNA-seq datasets, a subtle but consistent correlation between FAK and CD34 expression was seen ($p \leq 0.01$ for cell line data and q-value <0.0001, following Benjamini-Hochberg FDR correction, for both patient cohorts). These data suggest that blocking FAK may be most effective in $CD34^{high}$ expressing cases of AML (supplementary Figure 7a-c). In support of this, we found that defactinib sensitive KG1a cells had significantly higher protein expression of CD34 compared to OCI-AML3 cells (supplementary Table 3). It is worthy of note that adhered AML cells on the BMAS system also showed significantly higher CD34 expression than non-adhered (supplementary Figure 7d).

When defactinib was combined with 5µg/mL anti-CD44 treatment, the combination of the two prevented adhesion of KG1a cells up to a mean of 4.4-fold compared to untreated controls. This was substantially better than either drug alone, with the drugs combining in an additive fashion (mean Bliss Score=1.63) (Figure 6a and b). This indicates that dual targeting of these two pathways may represent a promising therapeutic strategy. To test this further, we co-cultured primary AML cells with autologous stromal cells grown out from the BM aspirate of 3 patients. The patient samples selected were known to have >15% $CD34^{high}$ AML cells in their BM aspirate. Similar to the cell line BMAS model, a significantly higher proportion of non-adhered $CD34^{+}$ cells were present in the samples treated with all concentrations of either anti-

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CD44 (up to 3.8-fold) or defactinib (up to 3.1-fold) alone. However, in samples treated with combinations of the two drugs, the number of non-adhered CD34+ cells were significantly higher than in the control or single drug-treated samples for all three patients and at all dose combinations tested. The most effective concentration combination was 2.5µg/mL anti-CD44 and 2.5µM defactinib for all three patients with changes of 14.2, 8.98 and 12.19-fold (compared to untreated), respectively. Representative scatter plots from one patient sample are shown in Figure 6b, showing substantially more non-adhered CD34+ and CD34- cells in the presence of both drugs, and combined data for the 3 patients in Figure 6c. Furthermore, the dose combinations were highly additive in all three patient samples at all concentrations (Bliss scores >1). At 2.5µg/mL of anti-CD44 + 2.5µM of defactinib the Bliss scores indicated synergy (Bliss score >10) in all patients (18.74, 11.59 and 12.93) (Figure 6d). Taken together these results suggest that defactinib and anti-CD44 treatment combine synergistically to reduce adhesion in patients with $CD34^{+}$ AML.

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Discussion

One of the issues hindering successful treatment of AML patients, and contributing to disease relapse, is leukemic cell adhesion and retention in the protective niche of the BMME. Here, the leukemic cells are surrounded by other cell types, including stromal, endothelial cells and osteoblasts, that promote their survival, by enabling them to evade destruction by both the immune system[38-42] and intra-vascular therapies[10,11], ultimately leading to the emergence of drug resistance. Disruption of these AML cell interactions in the BMME has potential to make them far more susceptible to standard therapies.

We have developed a novel, robust, and physiologically representative BM adhesion system, the BMAS. We used three cell types known to be abundant in the BM[7,8]; stromal cells (HS-5), endothelial cells and osteoblasts to create a highly adhesive system which we tested using both AML cell lines and primary cells. The system modeled the adhesiveness of the BM seen *in-vivo*[21] with a large percentage of AML cells adhered, more viable than their non-adhered counterparts. Furthermore, the BMAS was also shown to successfully recapitulate both CAM-DR and SFM-DR, again consistent with that seen in the AML BM[43,44]. Importantly, the system has utility as a drug testing platform.

To identify potentially druggable adhesion targets, we measured levels of adhesion molecules known to have commercially available blocking agents namely; CD44, CXCR4, CD49d and E-selectin and found all except E-selectin to be highly expressed on AML cells. In keeping with data from clinical trials[17,18], targeting CXCR4 with plerixafor reduced adhesion of a small proportion of AML cells in the BMAS and similar results were achieved using ONO-7161. Similarly, blocking CD49d with a clinical grade inhibitor commonly used for the treatment of multiple sclerosis[45], mobilized just

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a small number of AML cells. Even with the highest dose of anti-E-selectin, no effect was observed.

The potential importance of CD44 in AML adhesion and initiation has already been highlighted by Hartmann *et al*. [21]. Furthermore, anti-CD44 is available as a clinical grade humanized monoclonal antibody (RG7356) and has been trialled as a therapy for CD44-expressing solid tumours[46] and been found to be safe and well tolerated as a monotherapy in a phase 1 trial for AML[47]. However, its utilization as a BM anti-adhesion agent, to reverse CAM-DR and sensitize AML cells to chemotherapy, has not previously been tested. Here we show that blocking CD44 in our BMAS is effective in preventing AML adhesion and that combining this with cytarabine significantly increases AML cell apoptosis. Importantly, the combination of 5μM cytarabine and 5μg/mL anti-CD44 increased apoptosis of primary AML cells more than the additive effect of either alone, supporting the hypothesis that forcing AML cells out of the protective BM niche will render them more susceptible to standard therapies. This finding is supported in a study in by Bjorklund *et al*. who showed that blocking CD44 *in-vitro* reduced adhesion and sensitized myeloma cells to lenalidomide[25]. Furthermore, an *in-vivo* study utilized an anti-CD44 antibody to target liposomes containing cytarabine to AML cells. Although the study concluded that the reduced WBC in both BM and PB was due to the liposomal targeting, it is possible that blockade of CD44 and thus disruption of AML adhesion was also occurring[48].

Even though anti-CD44 treatment was the most effective blocking agent in our study, some AML cells still retained the ability to adhere. Transcriptional profiling of anti-CD44 treated persistently adhered AML cells identified them to be enriched in the FAK pathway and measurement of FAK phosphorylation confirmed increase FAK signalling indicating that inhibition of this in combination with anti-CD44 could be a promising

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therapeutic strategy. Defactinib (dual FAK and PYK2 inhibitor) is a clinically utilized drug that has been shown to have potential in models of chronic lymphocytic leukemia *in-vitro*[49]. In December 2022 (estimated completion 2025) a new phase I clinical trial started looking at the combination of defactinib and decitabine/cedazuridine (ASTX727) in AML, Myelodysplastic Syndrome and Chronic Myelomonocytic Leukaemia (CMML) patients (CELESTIAL-MDS trial). We showed that defactinib reduced adhesion in the more primitive KG1a CD34^{high} cell line and had an additive effect when combined with anti-CD44 treatment, preventing adhesion better than either agent alone. Interestingly, defactinib showed no effect in the more differentiated CD34^{low} expressing OCI-AML3 cells. Analysis of previously published CD34 and FAK (PTK2) mRNA levels in AML cell lines and two AML patient cohorts showed a subtle but consistent correlation between FAK and CD34[35-37] suggesting that defactinib might be most effective in more primitive CD34^{high} AML cells. Importantly, it is widely accepted that it is stem-like populations of cells that are associated with CAM-DR and relapse[50], and we show here that BMAS adhered AML cells have a more primitive CD34^{high} phenotype. The observation that defactinib preferentially targets these cells is a interesting finding of this study.

Importantly, the adhesion blockade seen in KG1a cells was replicated in CD34^{high} primary AML cells on an autologous BMME. The striking reduction in adhesion and synergy between the two drugs in all 3 of the primary samples tested, highlights the combination of anti-CD44 and defactinib as a promising and novel therapeutic mobilization strategy. As anti-CD44 is available as a recombinant immunoglobulin G1 humanized monoclonal antibody[46], and drug repositioning strategies always result in faster development to trial rates, addition of this agent to the defactinib and ASTX727 strategy has exciting implications for this largely incurable disease.

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Data availability

The datasets generated during and/or analysed during the current study are available in the EMBL-EBI data repository, accession number E-MTAB-14052.

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Author contributions

Eleni E. Ladikou designed the study, performed the research, analyzed the data and co-wrote the paper

Kim Sharp performed research

Fabio A. Simoes performed the research, analyzed the data and wrote the paper

John Jones provided essential samples and analyzed the data

Thomas Burley performed research

Lauren Stott performed research

Aimilia Vareli performed research

Emma Kennedy performed research

Sophie Vause performed research

Timothy Chevassut provided essential samples

Amarpreet Devi provided essential samples

Iona Ashworth performed research and provided essential samples

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David Ross analyzed data and co-wrote the paper

Tanja Nicole Hartmann analyzed data and co-wrote the paper

Simon Mitchell help design the study, analyzed data and co-wrote the paper

Chris Pepper help design the study, analyzed data and co-wrote the paper

Giles Best performed research, analyzed data and co-wrote the paper

Andrea G.S. Pepper designed the study, performed the research, analyzed the data and co-wrote the paper

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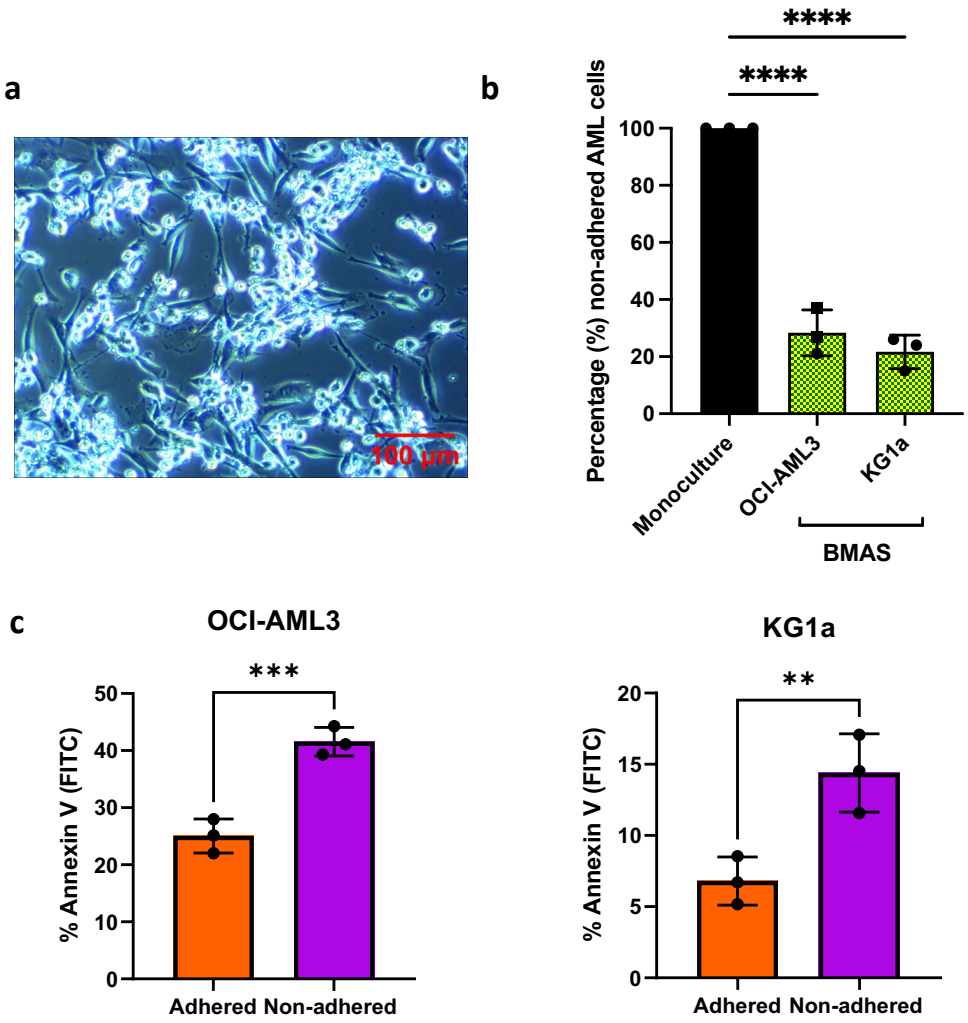


Figure 1: **The optimized BMAS.** (a) 10x magnification light microscopy images of co-culture. KG1a cells (bright round) attached to BMAS cells (darker and elongated). Images were captured using an Olympus CKX41 microscope, with a micropix camera and Tsview 7 version 7.1 software. (b) Percentage (%) non-adhered AML cells (mean±SD) when co-cultured with ratio of 1:1:1 hFOB1.19/HS-5/HUVEC; KG1a cells (n=3) and OCI-AML3 cells (n=3). Significance determined using a one-way ANOVA, following Shapiro–Wilk test for normality. (c) Percentage Annexin V positive (apoptotic) adhered versus non-adhered OCI-AML3 (n=3) and KG1a (n=3) cells. Significance determined by paired t-test, following Shapiro-Wilk test for normality. Significance determined by one sample t-test, following Shapiro-Wilk test for normality. **** $p \leq 0.0001$, *** $p \leq 0.001$, ** $p \leq 0.01$.

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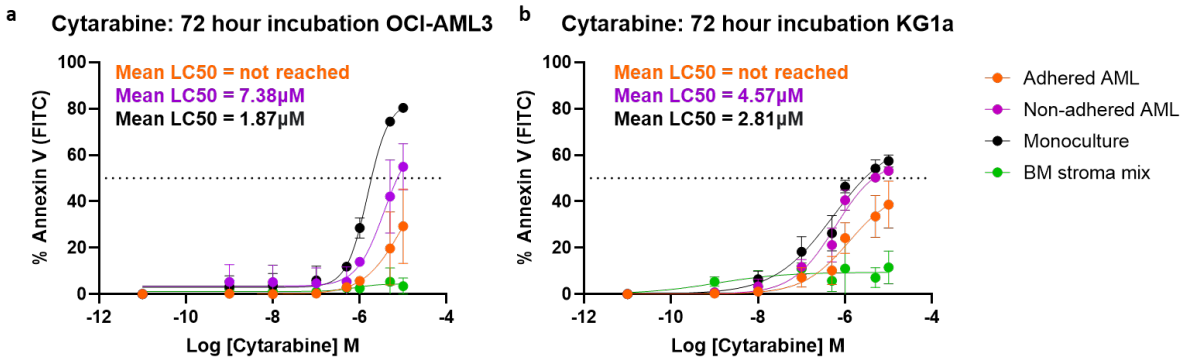


Figure 2: **BMAS modelled CAM-DR and SFM-DR following 72 hours of co-culture.** Cytarabine dose response curves (72 hours). (a) OCI-AML3 cells (n=3) and (b) KG1a cells (n=3) were treated with increasing doses of cytarabine for 72 hours and viability measured using Annexin V staining and flow cytometry. Sigmoidal dose response curves were plotted (mean±SD).

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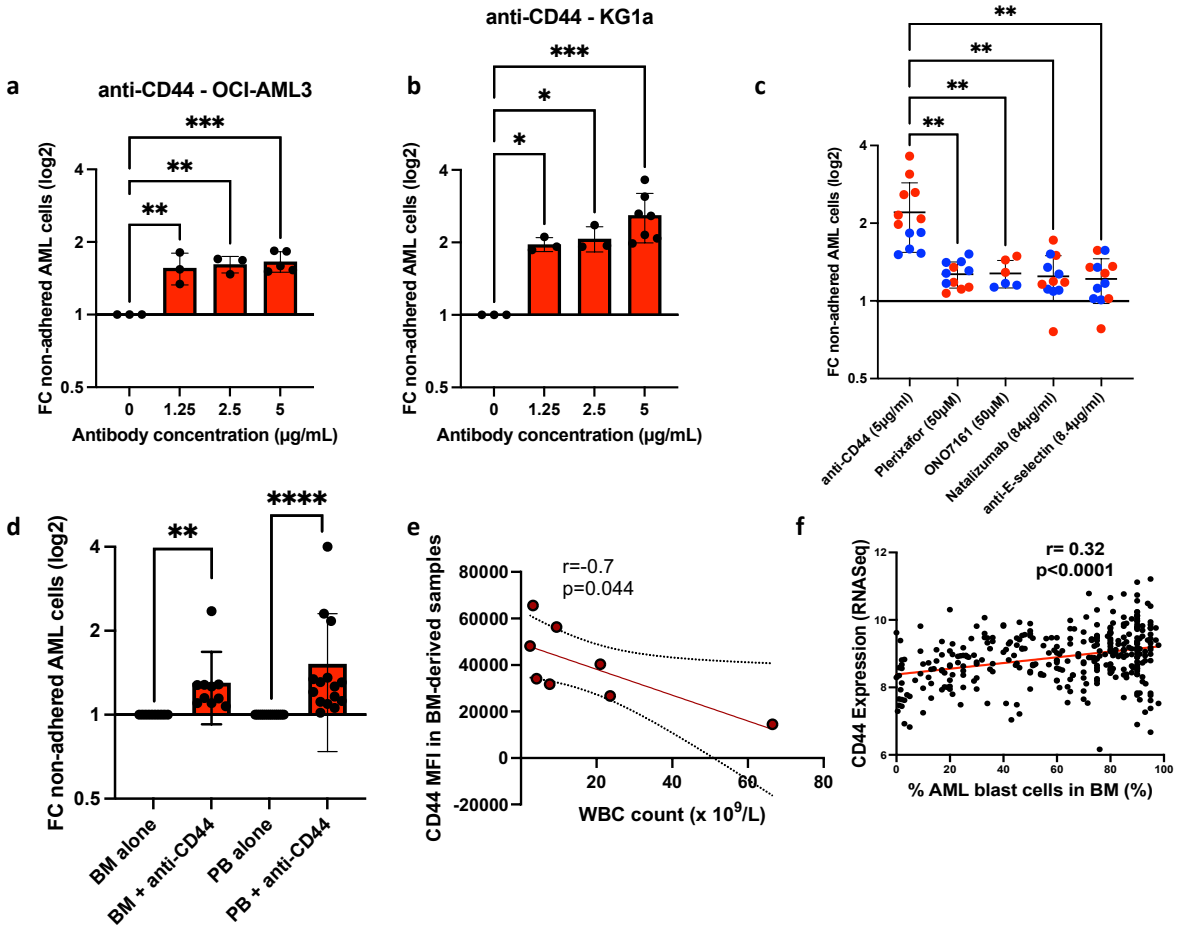


Figure 3: Anti-CD44 treatment was the most effective in blocking AML cell adhesion in cell lines and primary cells. Incubation of AML cells with anti-CD44 antibody. (a) Compared to untreated, fold change (FC) in non-adhered AML cells (mean±SD) when treated with increasing doses of anti-CD44 and co-cultured for 3 hours on the BMAS in (a) OCI-AML3 and (b) KG1a. Significance determined by one-way ANOVA and Dunnett's multiple comparisons test for comparing every mean to a no treatment control equal to 1. (c) A comparison between the best dose for each drug tested showing anti-CD44 was most effective (OCI-AML3 blue dots and KG1a red dots). Significance determined by Welch ANOVA with Dunnett's multiple comparisons. (d) Compared to untreated, fold change (FC) in non-adhered primary AML cells (mean±SD) when treated with 5µg/mL of anti-CD44 in BM (n=10) and PB (n=15) samples. Primary AML cells were identified using a full AML panel and patient-specific phenotyping data provided by the diagnostic laboratory. A representative panel and gating strategy for primary AML cells can be found in supplementary Figure 3. Significance determined by one sample

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Wilcoxon, following Shapiro-Wilk test for normality. Results are compared to a no treatment control equal to 1. ****p≤0.0001, **p≤0.01. (e) Correlation of CD44 expression (MFI) in BM-derived samples (n=8) with PB WBC count on samples taken at the same time. Correlation was determined using the Pearson's correlation with the 95% confidence interval shown as dotted lines. (f) Correlation of BM AML cell CD44 mRNA with the percentage of AML blast cells in the BM. Correlation was determined using the Pearson's correlation with the 95% confidence intervals.

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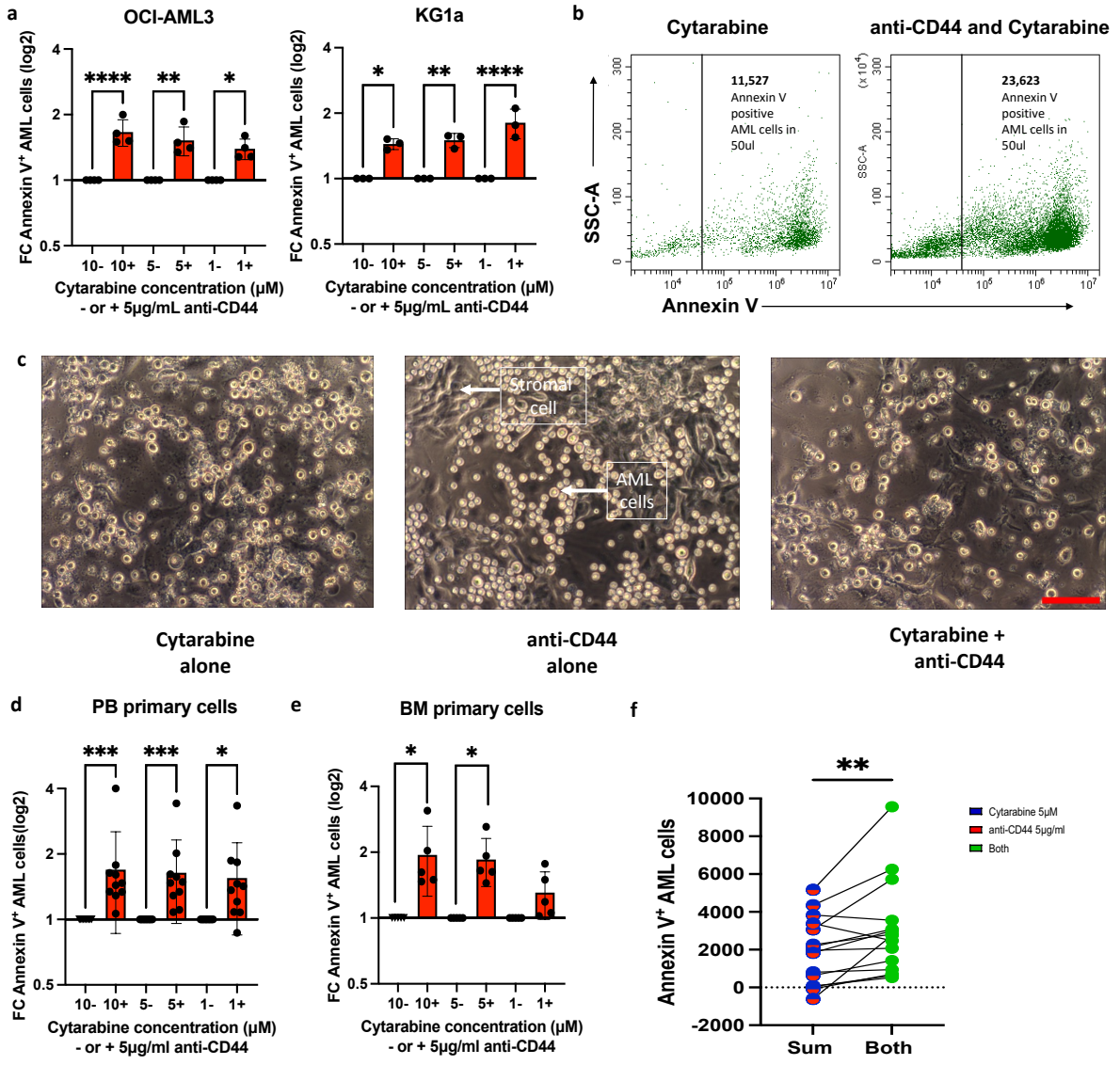


Figure 4: The combination of anti-CD44 with cytarabine can overcome CAM-DR. AML cells were incubated with three different concentrations of cytarabine ± pre-treatment with 5μg/mL anti-CD44 (a) For each cytarabine concentration (10μM, 5μM and 1μM), fold-change (FC) in the **number** of Annexin V positive AML cells (mean±SD) in the presence of anti-CD44 is compared to its absence (the absence of anti-CD44 is normalised to 1); OCI-AML3 (n=3) and KG1a cells (n=3). Significance determined by one-way ANOVA and Dunnett's multiple comparisons test comparing every mean to a no treatment control equal to 1 following Shapiro-Wilk test for normality (b) Representative dot plot of Annexin V staining of non-

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adhered KG1a cells following treatment with 1μM cytarabine alone (left) or 1μM cytarabine with 5μg/mL anti-CD44 (right). Although the proportion of apoptotic cells is similar in both, there are far more non-adhered KG1a cells in the presence of anti-CD44 (right) and therefore far more apoptotic KG1a cells. (c) Representative 10x magnification light microscopy images (scale bar represents 100μm) of the co-culture following treatment with 5μM cytarabine alone (left), 5μg/mL anti-CD44 alone (middle) or both (right). Elongated darker cells are the BMAS which adhered to the base of the well and are unaffected by chemotherapy and anti-CD44 treatment. Some persistently adhered AML cells (round with dark centre) were observed following cytarabine treatment alone (left image). More bright shiny AML cells are seen in the presence of anti-CD44 alone (middle image) but when cytarabine is added to anti-CD44, a marked reduction in the number of adhered AML cells was observed (right image). Images were captured using an Olympus CKX41microscpe, a micropix camera and Tsview 7 version 7.1 software. (d-e) For each cytarabine concentration (10μM, 5μM and 1μM), fold change (FC) in the **number** of Annexin V positive AML cells (mean±SD) in the presence of anti-CD44 is compared to its absence (the absence of anti-CD44 is normalised to 1) using (d) PB derived samples (n=10) and (e) BM derived samples (n=5). Significance was determined by Kruskal-Wallis test and Dunnett's multiple comparisons test (d) and one-way ANOVA test and Dunnett's multiple comparisons test (e), following Shapiro-Wilk test for normality. (f) Number of Annexin V positive AML cells (in 50ul) following treatment with 5μM of cytarabine alone added to that in 5μg/mL anti-CD44 alone in primary AML cells. The sum of their individual effect (red/blue column on left) is compared to their combination effect when cells were treated with both agents simultaneously (green column on right). Results for each individual patient is shown in supplementary Figure 4a). Each sample represents a biological repeat (n=15). Significance was determined by paired t-test, following Shapiro-Wilk test for normality. ****p≤0.0001, ***p≤0.001, **p≤0.01, *p≤0.05.

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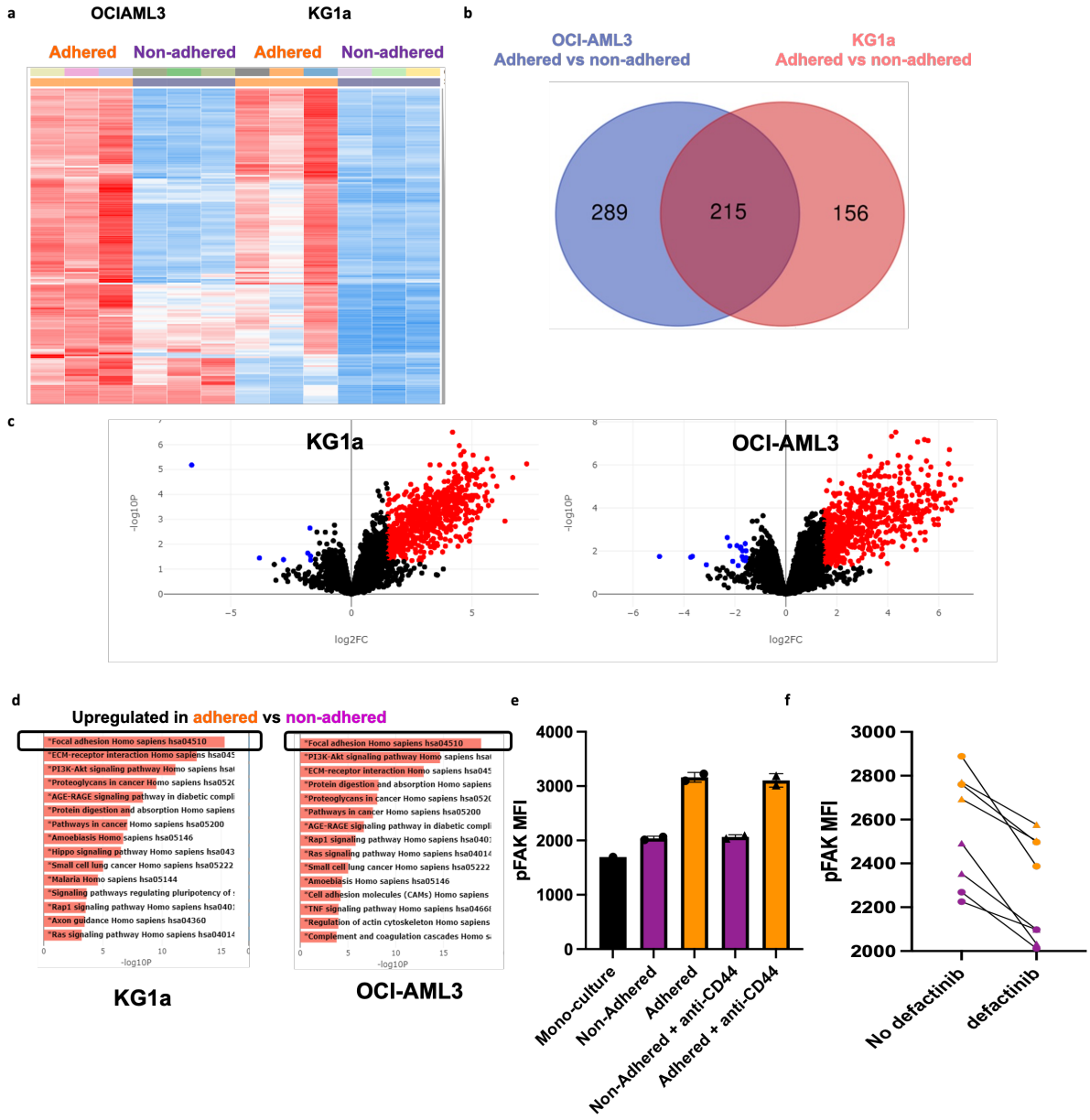


Figure 5: Transcriptomic analysis identified the FAK signaling pathways as the top determinant of persistent adhesion following treatment with anti-CD44. (a) Unsupervised hierarchical clustered heatmap for each sample in the RNA sequencing dataset. Every row of the heatmap represents a single gene, every column represents a sample, and every cell displays normalized gene expression values. (b) Venn diagram summarizing the overlap between differentially expressed genes between adhered versus non-adhered AML cells after each cell line was analysed separately. Left circle (blue) represents the genes differentially expressed in OCI-AML3 adhered cells compared non-adhered cells. Right circle (red)

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represents the genes differentially expressed in KG1a adhered cells compared to non-adhered cells. (c) Volcano Plot displaying the log2-fold changes of each gene calculated by performing a differential gene expression analysis. Every point in the plot represents a gene. Red points indicate significantly up-regulated genes, blue points indicate down-regulated genes for OCI-AML3 (right; 486 upregulated and 18 downregulated) and KG1a (left; 364 upregulated and 7 downregulated) cells. The thresholds used for this analysis were log2FC ≥1.5 and an adjusted P≤0.05. (d) Pathway Enrichment Analysis (KEGG pathways) for OCI-AML3 (right) and KG1a (left) cells. The x axis indicates the -log10(P-value) for each term. Significant terms are highlighted in bold. (e) MFI of pFAK in monoculture, adhered and non-adhered KG1a cells in the presence (triangle points) and absence (circle points) of 5μg/mL anti-CD44. (f) pFAK MFI, in adhered (orange triangles and circles) and non-adhered (purple triangles and circles) KG1a cells in the presence and absence of 5μM of defactinib.

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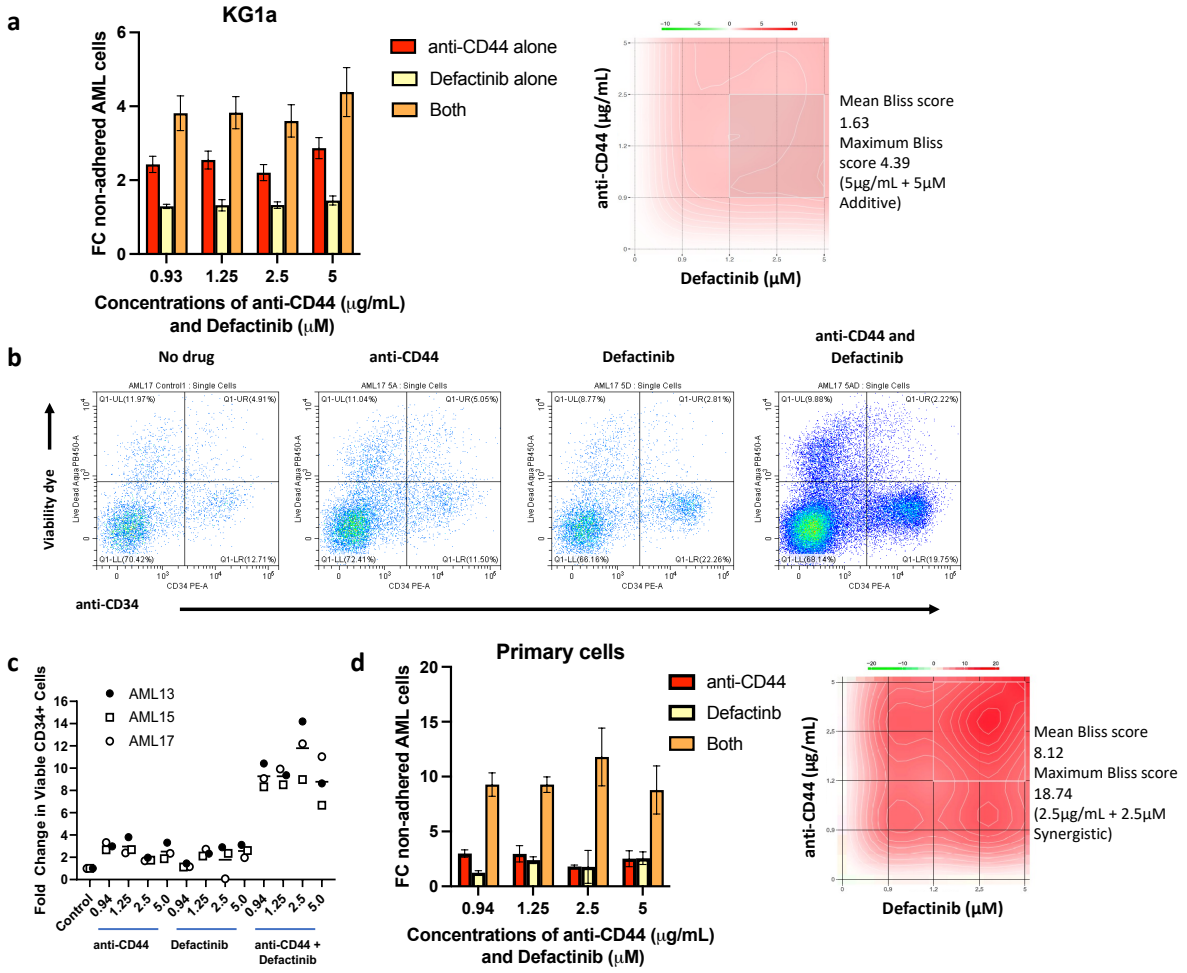


Figure 6: Defactinib in combination with anti-CD44 is additive/synergistic in preventing AML cell adhesion. (a) FC in non-adhered KG1a AML cells (mean±SD) when treated with increasing doses of anti-CD44 alone, increasing doses of defactinib alone or in combination. Synergy plot (right) was generated using SynergyFinder software (27) (https://synergyfinder.fimm.fi) shows an additive mean Bliss score of 1.627 (>1 =additive) and maximum of 4.39 (5μg/mL+5μM). Results are compared to no treatment control which is equal to 1. (b) Representative scatter plots of no drug, anti-CD44, defactinib and the combination of both showing the proportions of total and $CD34^+$ non-adhered primary AML cells after 2 minutes of acquisition on a Cytoflex S flow cytometer. This shows **substantially more** viable non-adhered $CD34^+$ and $CD34^-$ AML cells in the presence of both anti-CD44 and defactinib than no drug or either alone (c) Individual FC (compared to no drug) in viable **non-adhered**

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primary AML cell numbers (mean±SD) when treated with increasing doses of anti-CD44, defactinib or both and co-cultured for 3 hours with a confluent layer of autologous stromal cells. Different concentrations of anti-CD44 alone and defactinib alone versus the combination of both were determined by one-way ANOVA and the results are tabulated in supplementary Table 4; all comparisons were significant. (d) Combined FC in non-adhered primary AML cells (n=3, mean±SD) when treated with increasing doses of anti-CD44 alone, increasing doses of defactinib alone or in combination. A representative synergy contour plot for patient AML13 (right) was generated using SynergyFinder and shows a mean Bliss score of 8.12 and a synergistic maximum of 18.74 (2.5μg/mL+2.5μM).